

HELICOPTER PAD RECONSTRUCTION

SECTION 01 14 13 – SITE ACCESS, USE AND STORAGE

PART 1 - GENERAL

1.1 CONTRACTOR USE OF PROJECT SITE

- A. The Contractor's use of the project site shall include construction operations and storage of materials, and fabrication facilities, only within limit of work identified on the Drawings or as approved by the CO.
- B. It is the Contractor's responsibility to secure the site, materials, tools, equipment, and the work when not on site.

1.2 GOVERNMENT USE OF THE PROJECT SITE

- A. The Government may utilize the existing site during the period of construction for servicing the repeater equipment. The Contractor shall cooperate and coordinate with the Government to facilitate Government access and to minimize interference with the Contractor's operation at the same time. In any event, the Government shall be allowed access to the project site during the period of construction.

1.3 SUBMITTALS

- A. Submit the following items for approval a minimum of 5 calendar days prior to the Pre-Work conference. All plans shall be for the duration of the project.
 - 1. Camp Plan: Including but not limited to location and type of crew camp (field camp), facilities to be used, cooking facilities and storage, sanitation and water, and on-site power generation.
- B. The CO will review submittals within 14 calendar days then either approve plans or require changes. If required for approval, modify the submittals within 5 calendar days of receiving comments from the Government.
- C. The approved plans become part of the project. Both must be signed by the Contractor and the Government. A copy must be kept on site by the Contractor.

1.4 CONTRACTOR'S WORK CAMP AREA

- A. A land-based Work Camp location is contingent on the following requirements:
 - 1. For any land-based work camp location, obtain permission of CO for location, facilities to be included, and footprint (limit of disturbance) of Work Camp.
 - 2. Impacts from the land-based Work Camp shall be minimized to the existing site.
 - 3. The site shall be restored and all land-based camp facilities removed to the satisfaction of the CO.

HELICOPTER PAD RECONSTRUCTION

- B. Meet State of Alaska drinking water regulations 18 AAC 80 regarding potable water supplies for drinking and washing.
- C. Comply with State of Alaska wastewater disposal regulations 18 AAC 72 for domestic sewage.
- D. Ensure that the design, operation, and maintenance of all solid waste systems are protected from wildlife and meet all Federal, State, and local requirements. Meet State of Alaska solid waste management regulations (18 AAC 60) and Federal regulations contained in 40 CFR 243, 40 CFR 245, 40 CFR 257, and 40 CFR 258. Remove solid wastes and/or residues and dispose of in approved commercial landfills.
- E. Fuel storage shall meet the requirements of 40 CFR 112 (Oil Pollution Prevention).
- F. All food shall be stored in a location and manner as not to attract or allow access by bears and other wildlife.

1.5 LANDSCAPE PRESERVATION

- A. Operations shall be scheduled and conducted to minimize erosion of soils.
- B. Do not discharge pollutants such as raw sewage, and other harmful materials into or nearby streams, and impoundments or into natural or manmade channels leading thereto.
- C. Do not allow wash water or waste water from concrete or aggregate operations to enter ponds or live streams prior to treatment by filtration, settling, or other means sufficient to reduce the sediment content to not more than that of the water body into which it is discharged.
- D. Waste motor oil, fuel, lubricant, bitumen and similar hydrocarbons shall be collected and returned to the supplier. They shall not be wasted on the ground or buried.
- E. The subalpine muskeg at this site is fragile. Take all precautions necessary to protect it from damage. Contractor will be responsible for naturalizing any damaged areas as approved and at no expense to the Government.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 14 13

HELICOPTER PAD RECONSTRUCTION

SECTION 01 23 00 – OPTIONS

PART 1 - GENERAL

1.1 SUMMARY

- A. Option: An amount proposed by bidders and stated on the Bid Form for certain work defined in the bidding requirements that may be added to the base bid amount if the Government decides to accept a corresponding change either in the amount of construction to be completed or in the products, materials, equipment, systems, or installation methods described in the Contract Documents.
 - 1. Options described are part of the Work only if enumerated in the Agreement.
 - 2. The cost or credit for each Option is the net addition to the Contract Sum to incorporate alternate into the Work. No other adjustments are made to the Contract Sum.

1.2 PROCEDURES

- A. Coordination: Revise or adjust affected adjacent work as necessary to completely integrate work of the option into Project.
 - 1. Include as part of each option, miscellaneous devices, accessory objects, and similar items incidental to or required for a complete project whether or not indicated as part of option.
- B. Notification: Immediately following award of the Contract, the CO shall notify the Contractor, in writing, of the status of each option. Indicate if options have been accepted, rejected, or deferred for later consideration. Include a complete description of negotiated revisions to options.
- C. Execute accepted options under the same conditions as other Work of the Contract.

1.3 SCHEDULE OF OPTIONS

- A. Descriptions of the options are found in the Schedule of Items and Contract Documents.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 23 00

HELICOPTER PAD RECONSTRUCTION

SECTION 01 33 00 – SUBMITTALS

PART 1 - GENERAL

1.1 GENERAL

- A. Wherever submittals are required hereunder, all such submittals by the Contractor shall be submitted to the CO.
- B. Within 14 days after the date of Notice to Proceed as stated in the Notice to Award, the Contractor shall submit the following items to the CO for review:
 - 1. A preliminary schedule of Construction Drawings, Samples, and proposed Substitutes or "Or-Equal" submittals.
 - 2. A list of all permits and licenses the Contractor shall obtain indicating the agency required to grant the permit and the expected date of submittal for the permit and required date for receipt of the permit.
 - 3. A complete progress schedule for all phases of the Project.
 - 4. All required Material Safety Data Sheets.
 - 5. A letter designating the Contractor's Superintendent, defining that person's responsibility and authority.
 - 6. All Operations and Work Coordination Plans as indicated in Site Access, Work, and Storage Sections and elsewhere in the Specifications. These include, but are not limited to:
 - a. Camp Plan
 - 7. Qualifications as indicated in the Specifications. These include, but are not limited to:
 - a. Project Supervisor/Superintendent
 - b. Welder Supervisor
 - c. Professional Engineer Certification
- C. Within 90 days or deemed acceptable duration by Contractor and CO, after the date of commencement as stated in the Notice to Award, the Contractor shall submit the following items to the CO for review:
 - 1. Delegated Design Submittals: Signed and stamped Engineering drawings and calculations for Delegated Design/Performance Based items:
 - a. Helicopter pad and foundation. These include, but are not limited to:
 - 1) Structural design values
 - 2) Structural calculations
 - 3) Final 100% Construction Drawings based on government furnished 90% draft drawings (Sheets 2 – 6).

HELICOPTER PAD RECONSTRUCTION

- a) Interim review period: Provide Government 21 calendar days for review and comments of 95% drawings prior to finalizing and stamping final submittal.
 - b) Incorporate Government comments into final submittal.
 - 4) Manufacturer's specification sheets for all products and materials including aluminum, telescoping jacks and concrete.
- D. No payments shall be made to the Contractor until all these items are submitted in their entirety, as determined by the CO.
- E. All proposed variations shall conform to applicable building codes.
- F. Construction drawings shall bear the stamp of an Alaska registered Professional Engineer. Calculations shall be prepared and stamped by a Professional Engineer licensed in the state of Alaska.
- G. Special Project Specifications provided by the government associated with construction of the helicopter pad and associated appurtenances are for design reference only unless specifically incorporated into contractor's stamped drawings and specifications, and approved by CO. Approved construction drawings and specifications shall take precedence over preliminary design drawings and specifications provided by the Government.
- H. The Government shall review the submittal and within 21 calendar days provide a written response to the contractor outlining any corrections to the proposed working drawings. The Contractor shall then have 14 calendar days to submit the corrected shop drawings to the Contracting Officer for approval. Approval of the shop drawings will constitute the Contractor's Notice to Proceed.

1.2 SUBMITTALS

- A. Wherever called for in the Contract Documents, or where required by the CO, the Contractor shall furnish to the CO, for review, one digital copy of each submittal as a pdf.
- B. All Drawing submittals shall be accompanied by the Contractor's standard submittal transmittal form. Any submittal not accompanied by such a form, or where all applicable items on the form are not completed, will be returned for re-submittal.
- C. Normally, a separate transmittal form shall be used for each specific item or class of material or equipment for which a submittal is required. Transmittal of a submittal of various items using a single transmittal form will be permitted only when the items taken together constitute a manufacturer's "package" or are so functionally related that expediency indicates review of the group or package as a whole. A multiple-page submittal shall be collated into sets, and each set shall be stapled or bound, as appropriate, prior to transmittal to the CO.
- D. Except as may otherwise be provided herein, the CO will respond and return one digital copy of each submittal to the Contractor within 14 calendar days following their

HELICOPTER PAD RECONSTRUCTION

receipt by the CO. It is considered reasonable that the Contractor shall make a complete and acceptable second re-submittal to the CO if the initial submittal is rejected or deemed unacceptable. The Government reserves the right to withhold monies due the Contractor to cover additional costs of the CO review beyond the second submittal. The CO's maximum review period for each submittal including all re-submittals will be 14 days per submission. In other words, for a submittal that requires 2 re-submittals before it is complete, the maximum review period for that submittal could be 28 days.

- E. If submittal is returned to the Contractor marked "NO EXCEPTIONS TAKEN," formal revision and re-submission of said submittal will not be required.
- F. If submittal is returned to the Contractor marked "MAKE CORRECTIONS NOTED," formal revision and re-submission of said submittal will not be required.
- G. If submittal is returned to the Contractor marked "AMEND-RESUBMIT," the Contractor shall revise said submittal and shall resubmit the required number of copies of said revised submittal to the CO.
- H. If submittal is returned to the Contractor marked "REJECTED-RESUBMIT," the Contractor shall revise said submittal and shall resubmit the required number of copies of said revised submittal to the CO.
- I. Fabrication of an item may be commenced only after the CO has reviewed the pertinent submittal and returned copies to the Contractor marked either "NO EXCEPTIONS TAKEN" or "MAKE CORRECTIONS NOTED." Corrections indicated on submittal shall be considered as changes necessary to meet the requirements of the Contract Documents and shall not be taken as the basis for changes to the Contract requirements. Only a modification signed by the CO can alter the contract price, time, or requirements.
- J. All Contractor submittals shall be carefully reviewed by an authorized representative of the Contractor, prior to submission to the CO. Each submittal shall be dated, signed, and certified by the Contractor, as being correct and in strict conformance with the Contract Documents. In the case of shop drawings, each sheet shall be so dated, signed, and certified. No consideration for review by the CO of any Contractor submittal will be made for any items which have not been so certified by the Contractor. All non-certified submittals will be returned to the Contractor without action taken by the CO, and any delays caused thereby shall be the total responsibility of the Contractor.
- K. The CO's review of Contractor shop drawing submittals shall not relieve the Contractor of the entire responsibility for the correctness of details and dimensions. The Contractor shall assume all responsibility and risk for any misfits due to any errors in Contractor submittals. The Contractor shall be responsible for the dimensions and the design of adequate connections and details.

HELICOPTER PAD RECONSTRUCTION

1.3 RECORD DRAWINGS SUBMITTAL

- A. The Contractor shall keep and maintain, at the job site, one record set of Drawings. On these, it shall mark all Project conditions, locations, configurations, and any other changes or deviations which may vary from the details represented on the original Contract Drawings, including buried or concealed construction and utility features which are revealed during the course of construction. Special attention shall be given to recording the horizontal and vertical location of all buried construction that differ from the locations indicated, or which were not indicated on the Contract Drawings. Said record drawings shall be supplemented by any detailed sketches as necessary or directed to indicate, fully, the Work as actually constructed. These master record drawings of the Contractor's representation of as-built conditions, including all revisions made necessary by addenda, modifications, and the like shall be maintained up-to-date during the progress of the Work.
- B. In the case of those drawings which depict the detail requirement for equipment to be assembled in the factory, such as the spiral staircase and the like, the record drawings shall be updated by indicating those portions which are superseded by modification drawings or final shop drawings, and by including appropriate reference information describing the modifications by number and the shop drawings by manufacturer, drawing, and revision numbers.
- C. Record drawings shall be accessible to the CO at all times during the construction period and shall be delivered to the CO on the 20th working day of every third month after the month in which the Notice to Proceed is given as well as upon completion of the Work.
- D. Final payment will not be acted upon until the Contractor-prepared record drawings have been delivered to the CO.

1.4 PROGRESS SCHEDULES

- A. The progress schedule shall be in Bar Chart or Critical Path Method (CPM) form, as required by the CO.
- B. The progress schedule shall show the order in which the Contractor proposes to carry out the Work and the completed date on which the Contractor and their subcontractors will start and finish each of the salient features of the Work, including any scheduled periods of shutdown. The schedule shall also indicate any anticipated periods of multiple-shift work.
- C. Upon substantial changes to the Contractor's progress schedule of Work or upon request of the CO, the Contractor shall submit a revised progress schedule(s) in the form required. Such revised schedule(s) shall conform with the Contract Time and take into account delays which may have been encountered in the performance of the Work. In submitting a revised schedule, the Contractor shall state specifically the reason for the revision and the adjustments made in its schedule or methods of operation to ensure the completion of all the Work within the contract time.

HELICOPTER PAD RECONSTRUCTION

1.5 PROPOSED SUBSTITUTES OR "OR-EQUAL" ITEM SUBMITTAL

- A. Whenever materials or equipment are specified or described in the Contract Documents by using the name of a proprietary item or the name of a particular Supplier, the naming of the item is intended to establish the type, function, and quality required. If the name is followed by the words "or-equal" indicating that a substitution is permitted, materials or equipment of other Suppliers may be accepted by the CO if sufficient information is submitted by the Contractor to allow the CO to determine that the material or equipment proposed is equivalent or equal to that named, subject to the following requirements:
1. The burden of proof as to the type, function, and quality of any such substitute material or equipment shall be upon the Contractor.
 2. The CO will be the sole judge as to the type, function, and quality of any such substitute material or equipment and the CO's decision shall be final.
 3. The CO may require the Contractor, to furnish at the Contractor's expense, additional data about the proposed substitute.
 4. The CO may require the Contractor to furnish at the Contractor's expense a special performance guarantee or other surety with respect to any substitute.
 5. Acceptance by the CO of a substitute item proposed by the Contractor shall not relieve the Contractor of the responsibility for full compliance with the Contract Documents and for adequacy of the substitute item.
 6. The Contractor shall be responsible for resultant changes and all additional costs which the accepted substitution requires in the Contractor's Work, the Work of its subcontractors and of other contractors, and shall effect such changes without cost to the Government. This shall include the cost for redesign and claims of other contractor affected by the resulting change.
- B. The procedure for review by the CO will include the following:
1. If the Contractor wishes to furnish or use a substitute item of material or equipment, the Contractor shall make written application to the CO for acceptance thereof. The substitution application must meet the criteria set in paragraphs C and D below.
 2. Unless otherwise provided by law or authorized in writing by the CO, the "Substitution Requests" shall be submitted within the 21-day period after Notice of Award.
 3. Wherever a proposed substitute material or equipment has not been submitted within said 21-day period, or wherever the submission of a proposed substitute material or equipment has been judged to be unacceptable by the CO, the Contractor shall provide material or equipment named in the Contract Documents.
 4. The Contractor shall certify that the proposed substitute will perform adequately the functions and achieve the results called for by the general design, be similar and of equal substance to that specified, and be suited to the same use as that specified.
 5. The CO will be allowed a reasonable time within which to evaluate each proposed substitute. In no case will this reasonable time period be less than 14 days.

HELICOPTER PAD RECONSTRUCTION

6. As applicable, no shop drawing submittals will be made for a substitute item nor will any substitute item be ordered, installed, or utilized without the CO's prior written acceptance of the Contractor's "Substitution Request".
- C. The Contractor's application using the "Substitution Request" shall contain the following statements and/or information which shall be considered by the CO in evaluating the proposed substitution:
1. The evaluation and acceptance of the proposed substitute will not impact the Contractor's achievement of substantial completion on time.
 2. Whether or not acceptance of the substitute for use in the Work will require a change in any of the Contract Documents to adopt the design to the proposed substitute.
 3. Whether or not incorporation or use of the substitute in connection with the Work is subject to payment of any license fee or royalty.
 4. All variations of the proposed substitute for that specified will be identified.
 5. Available maintenance, repair, and replacement service and its estimated cost will be indicated.
 6. Itemized estimate of all costs that will result directly or indirectly from acceptance of such substitute, including cost of redesign and claims of other contractors affected by the resulting change.
- D. Submit separate requests for each substitution. Support each request with the following:
1. Complete data substantiating compliance of proposed substitution item(s) with the requirements stated in contract documents:
 - a. Product identification, including manufacturer's name and address;
 - b. Manufacturer's literature, identifying:
 - 1) Product description;
 - 2) Reference standards, and;
 - 3) Performance and test data.
 - c. Samples, as applicable (see AGAR 452.236-76);
 - d. Name and address of similar projects on which the substitution product has been used, and date of the installation;
 - e. Comparative data of item(s) to be deleted through substitution. Include drawing to the same (or larger) scale as pertinent portion of contract documents and marked to show how differences will be accommodated;
 - f. Show the complete system/assembly as revised to accommodate the proposed alternative for difference in performance, size, configuration, connection(s) service, accessibility, or any other signification characteristics;
 - g. Show the complete layout of the system except that which is identical to contract documents, unless unchanged portion must be shown to indicate clearances, etc.;
 - h. Include complete detail drawings of supports for all substitute equipment, and complete load calculations for adequacy of support prepared and signed by a registered structural engineer.

HELICOPTER PAD RECONSTRUCTION

2. Itemized comparison of the proposed substitution with product specified; list significant variations.
3. Data relating to changes in construction schedule.
4. Any effect of substitution on separate contracts.
5. List of changes required in other work or products.
6. Accurate cost data comparing proposed substitution with product specified.
7. Proposed amount of change to the contract value.
8. Designation of required license fees or royalties.
9. Designation of availability of maintenance services, and/or sources of replacement materials.
10. Substitutions will not be considered for acceptance when:
 - a. They are merely indicated or implied on shop drawings, or product data submittals without a formal request received from the Contractor;
 - b. They are requested directly by a subcontractor or supplier;
 - c. Acceptance will require substantial revision of contract documents, or;
 - d. Substitution has not been requested in a way this and related contract clauses.
11. Substitute products must not be ordered or installed without written acceptance by the CO.
12. The CO will determine acceptability of proposed substitutions in consultation with other Government employees with knowledge of the project and/or product(s).
13. Number of Substitution Submittals: only one proposed alternative will be considered for each substituted item. The Contractor must provide the originally-specified item(s) if the substitution proposal is rejected.

1.6 CONTRACTOR SUBSTITUTION REQUEST REPRESENTATION AND RESPONSIBILITIES

- A. In making a formal request for substitution of item(s), the Contractor represents that:
1. Have investigated the product proposed to substitute the originally-specified item, and have determined that it is either equal to, or superior to, the original item in all respects to that originally-specified item;
 2. Will provide the same or better warranties and bond(s) for the substitute as for the originally-specified item;
 3. Will coordinate the installation of any accepted substitution into the work, and will make such changes as may be required for the work to be completed in all respects;
 4. Will waive any and all claims for additional costs caused by substitution, which may not be readily apparent, and may become apparent only subsequently, and;
 5. Cost data is complete and includes related costs under the contract, other than:
 - a. Costs under separate contracts, and;
 - b. CO costs for redesign or revision of contract documents.
 6. The Contractor assumes full responsibility that the substitution and/or alternate item(s) or procedure(s) will meet the job requirement(s).

HELICOPTER PAD RECONSTRUCTION

7. The Contractor is solely responsible for the cost of redesign and modification(s) to this and other parts of the work caused by the substitution(s) or alternate item(s) furnished.

PART 2 - PART 2 - PRODUCTS (Not Used)

PART 3 - PART 3 - EXECUTION (Not Used)

END OF SECTION 01 33 00

HELICOPTER PAD RECONSTRUCTION

SECTION 01 42 00 – REFERENCES AND DEFINITIONS

PART 1 - GENERAL

1.1 GENERAL

- A. Titles of Sections and Paragraphs. Captions accompanying Specification sections and paragraphs are for convenience of reference only, and do not form a part of the Specifications.
- B. Applicable Publications. Whenever in these Specifications references are made to published specifications, codes, standards, or other requirements, it shall be understood that wherever no date is specified, only the latest specifications, standards, or requirements of the respective issuing agencies which have been published as of the date that the Work is advertised for bids, shall apply; except to the extent that said standards or requirements may be in conflict with applicable laws, ordinances, or governing codes. No requirements set forth herein or shown on the Drawings shall be waived because of any provision of, or omission from, said standards or requirements.

1.2 REFERENCE SPECIFICATIONS, CODES, AND STANDARDS

- A. Without limiting the generality of other requirements of the Specifications, all Work specified herein shall conform to or exceed the requirements of applicable codes and the applicable requirements of the following documents.
- B. References herein to "Building Code" or "International Building Code" shall mean International Building Code of the International Code Council.
- C. The Contractor shall construct the Work specified herein in accordance with the requirements of the Contract Documents and the referenced portions of those referenced codes, standards, and specifications listed herein.
- D. References herein to "OSHA Regulations for Construction" shall mean Title 29, Part 1926, Construction Safety and Health Regulations, Code of Federal Regulations (OSHA), including all changes and amendments thereto.
- E. References herein to "OSHA Standards" shall mean Title 29, Part 1910, Occupational Safety and Health Standards, Code of Federal Regulations (OSHA), including all changes and amendments thereto.
- F. The technical specifications, standards, and guidelines referenced elsewhere in the Specifications and Drawings and listed below are hereby incorporated into the Work.
 - 1. The Federal Highways Administration Standard Specifications for Construction of Roads and Bridges on Federal Highway Projects, FP-14 (abbreviated as FP-14), are included by reference. An electronic copy is available upon request.

HELICOPTER PAD RECONSTRUCTION

2. Forest Service Handbook Alaska Region (R10) Juneau, AK. FSH2509.22 Soil and Water Conservation Handbook for National Forest System Lands in Alaska. R-10 Amendment 2509.22-2006-2.
 3. USDA Forest Service National Best Management Practices (BMPs) for Water Quality Management on National Forest System Lands. Vol 1: National Core BMP Technical Guide, FS-990a.
 4. Forest Service Supplements to FP-14
 5. Sign and Poster Guideline for the Forest Service EM 7100-15 (Revised October 2013)
 6. Any project related permits obtained by the Government or Contractor and the conditions and restrictions contained within.
- G. In case of conflict between codes, reference standards, Drawings and the other Contract Documents, the most stringent requirements shall govern. All conflicts shall be brought to the attention of the CO for clarification and directions prior to ordering or providing any materials or furnishing labor. The Contractor shall bid for the most stringent requirements.

1.3 DEFINITIONS APPLICABLE TO TECHNICAL SPECIFICATIONS

- A. The following words have the meaning defined in the Technical Portions of the Work:
1. Construction Documents: The Drawings and technical specifications that describe the Work and are hereby made part of this solicitation and any resultant contract.
 2. Contracting Officer (CO): Representative of the Government overseeing the project.
 3. Demolish: Typically through mechanical means, the complete removal of existing construction or existing features and legally dispose of off-site.
 4. Existing to Remain/Retain/Protect: Existing items or features that are not to be removed and that are not otherwise indicated to be removed, removed and salvaged, or removed and reinstalled.
 5. Furnish: To supply and deliver to the site, to unload and unpack ready for assembly, installation, testing, and start-up.
 6. Government: The USDA Forest Service, Owner.
 7. Indicated: A word used to direct the Contractor to information contained on the drawings or in the Specifications. Terms such as "shown", "noted", "scheduled", and "specified" also may be used to assist in locating information but no limitation of location is implied or intended.
 8. Install: Assembly, erection, placing, anchoring, applying, shaping to dimension, finishing, curing, protecting, and cleaning, ready for the Owner's use.

HELICOPTER PAD RECONSTRUCTION

9. Installer: A person or firm engaged by the Contractor or its subcontract, or any subcontractor, for the performance of installation, erection, or application Work at the site. Installers must be expert in the operations they are engaged to perform.
10. Material in Kind: Material that matches existing materials, as much as possible, in species, cut, color, grain, size, and finish.
11. Provide: To furnish and install, ready for the intended use.
12. Remove: Detach items from existing construction or removal of existing features and legally dispose of them off-site, unless indicated to be removed and salvaged or removed and reinstalled.
13. Remove and Salvage: Detach items from existing construction and deliver them to Owner ready for reuse.
14. Remove and Reinstall: Detach items from existing construction or entire existing features or structures, clean, repair, prepare for reuse, and reinstall where indicated.
15. Repair: To stabilize, consolidate, or conserve; to retain existing materials and features while employing as little new material as possible. Repair includes patching, piecing-in, splicing, consolidating, or otherwise reinforcing or upgrading materials. Within restoration, repair also includes limited replacement in kind, rehabilitation, and reconstruction, with compatible substitute materials for deteriorated or missing parts of features when there are surviving prototypes.
16. Replace: To duplicate and replace entire features with new material in kind.
17. Stockpile: To remove, accumulate, and protect on site for future use. Typically reinstalled, unless otherwise noted.

1.4 TECHNICAL SPECIFICATIONS

- A. These technical specifications are hereby made part of this solicitation and any resultant contract.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 42 00

HELICOPTER PAD RECONSTRUCTION

SECTION 02 41 00 – DEMOLITION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Includes all labor, materials and equipment for demolition, removal and disposal of existing helicopter pad structure, footings, cable ties and any associated accessories.

1.3 MATERIALS OWNERSHIP

- A. Unless otherwise indicated, demolition waste becomes property of the Contractor.

1.4 SUBMITTALS

- A. Delegated Design Submittal: Helicopter Pad Structure. Provide a completed set of plans and calculation package stamped by a registered professional engineer licensed in the State of Alaska. The submittal shall include a stamped set of calculations to justify design meets performance requirements, loading requirements identified in Section 132110 and all applicable building code requirements.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 DEMOLITION, GENERAL

- A. Carry out all demolition work in a neat and orderly manner. Keep noise, dust, and similar nuisances to a minimum. Do not throw or drop materials.
- B. Legally dispose of demolition materials off site of National Forest System Lands, and in accordance with all Federal, State, and Local laws and regulations.
- C. Within heavy traffic areas adjacent to helicopter pad footprint, protect low-growing groundcover vegetation.

HELICOPTER PAD RECONSTRUCTION

- D. Remove taller vegetative materials over 18 inches high for an area of 60 feet by 60 feet for landing zone.

3.2 ITEMS FOR REMOVAL/DEMOLITION

- A. Items to be removed/demolished include, but are not limited to, the following:
 - 1. Helicopter Pad and related support structures, concrete footings, anchoring materials, and other minor related structures. Cut foundation posts at ground level. Disassemble framing members and break up above-grade concrete if any, as required for complete removal.
- B. All demolished items and materials to be disposed of shall become the property of the Contractor. Remove and dispose of all demolished items and demolition materials off-site in accordance with all local, state, and federal laws, rules, and regulations.

3.3 ITEMS FOR PROTECTION AND PRESERVATION

- A. Items to be protected and preserved include, but are not limited to, the following:
 - 1. Radio Repeater equipment
 - 2. Comshel Shelter

END OF SECTION 02 41 00

HELICOPTER PAD RECONSTRUCTION

SECTION 09 93 00 – PAINTS, STAINS AND TRANSPARENT FINISHES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes primer and paints for metal.

1.3 SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Product Samples: For each type of product indicated.
- C. Sample of each paint, stain and finish on each specified material appropriately finished. Minimum size: 2-inch by 6-inch or an alternate size approved by CO.

1.4 DELIVERY, STORAGE & HANDLING

- A. Storage and Protection: Store materials protected from exposure to harmful environmental conditions and at temperature and humidity conditions recommended by the manufacturer.
- B. Delivery and Storage of Materials: Deliver in unopened containers; manufacturer's original labels thereon. Store in dry location, protected from weather.

PART 2 - PRODUCTS

2.1 STEEL FINISH

- A. Provide the following finishes, or approved equal.
- B. For the Telescoping Steel Jack Post
 - 1. First Coat: POR 15 Rust Preventative
 - 2. Top Coat: POR-15 Top Coat
 - a. Color: Silver

HELICOPTER PAD RECONSTRUCTION

PART 3 - EXECUTION

3.1 GENERAL

- A. Workmanship: Highest quality by skilled workmen to CO satisfaction. Apply material evenly without runs, thin spots, skips or other finishing and visual defects. Finish surfaces uniformly, free from streaks, blotches and other defects.
- B. Prepare material and apply as directed by manufacturer.

3.2 ALUMINUM FINISH APPLICATION

- A. All finishes shall be shop applied as directed by paint manufacturer.
- B. Metal Preparation: As directed by paint manufacturer.
- C. Application: Apply as directed by manufacturer.
 - 1. Primer: Minimum dry film thickness of 3mils.
 - 2. Finish: Minimum dry film thickness of 3mils.
- D. Tape all edges between painted and unpainted surfaces. Edges to be clean and sharp without overspray or paint on surfaced to remain unpainted.
- E. All damaged surfaces shall be repaired with the two coat system and surfaces shall be prepared per the manufacturer's recommendations for field application of the paint system. CO to approve all damaged surfaces and field applied paint systems.
- F. Clean all surfaces as recommended by manufacturer.

END OF SECTION 09 93 00

HELICOPTER PAD RECONSTRUCTION

SECTION 13 21 10 – HELICOPTER PAD DELEGATED DESIGN AND CONSTRUCTION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Delegated design for the helicopter pad structure, including but not limited to all framing members, aluminum stairs, concrete foundation, cable ties, and anchor bolts.
- B. Related Requirements:
 - 1. General Notes and details as shown on the Plans.

1.3 DEFINITIONS

- A. Basis of Design is the 90% design of USFS Helicopter Pad Reconstruction sheets 2 – 6 provided in the drawing set.

1.4 ACTION SUBMITTALS

- A. Delegated Design Submittal: For Helicopter Pad shown on the plans, provide set of plans with updated construction notes and supplemental specification as needed and calculation package stamped by a registered professional engineer licensed in the State of Alaska. Plans shall include enough information for all aspects of construction so that the Contractor can construct a complete pad.
 - 1. The submittal shall include a stamped set of calculations to justify design meets performance requirements for projected snow load at the most extreme provided locations, and loading requirements identified in the FAA Advisory Circular 150/5390-2C for Heliport Design and referencing the NWCG Standards for Helicopter Operations (NSHO) for both A-Star and Bell 407 helicopters for design, static, dynamic and rotor loads.
 - 2. Submittal shall have Government interim review at 95% complete prior to final stamped submittal. This allows Government review of design. See Definitions, Products, and Execution for basis of design for helicopter pad.

HELICOPTER PAD RECONSTRUCTION

1.5 INFORMATIONAL SUBMITTALS

- A. Welding Certificates.

1.6 QUALITY ASSURANCE

- A. Welding Qualifications: Qualify procedures and personnel according to the following:
 - 1. AWS D1.1/D1.1M, "Structural Welding Code - Steel."
 - 2. AWS D1.2/D1.2M, "Structural Welding Code - Aluminum."

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Schedule delivery of materials to avoid extended on-site storage and to avoid delaying the Work.
- B. Store materials under cover and protected from weather and contact with damp or wet surfaces. Provide for air circulation within and around stacks and under temporary coverings.

PART 2 - PRODUCTS

2.1 HELICOPTER PAD MATERIALS, GENERAL

- A. For materials not defined in this specification, the Construction Notes and Supplemental Specifications shall be referenced.

2.2 PERFORMANCE REQUIREMENTS

- A. Delegated Design: Engage a qualified registered professional engineer licensed in the State of Alaska to provide calculations package and final construction plans.
- B. Structural Performance: The helicopter pad shall withstand the effects of gravity loads and the following loads and stresses within limits and under conditions indicated.
 - 1. Site specific design requirements for most extreme locations shown on the plans including self-weight, snow loads, A-Star and Bell 407 helicopter design loads, static loads, dynamic loads, and rotor loads.
 - 2. Reference the design load requirements in FAA Advisory Circular 150/5390-2C for Heliport Design and the NWCG Standards for Helicopter Operations (NSHO).
- C. Lateral Performance: Helicopter pad shall withstand the effects of lateral loads as determined by ASCE 7 and as follows:
 - 1. Site specific design criteria for most extreme locations shown on the plans for both wind and seismic forces.

HELICOPTER PAD RECONSTRUCTION

2.3 MATERIALS

- A. Metal Surfaces, General: Provide materials with smooth surfaces, without seam marks, roller marks, rolled trade names, stains, discolorations, or blemishes.
- B. Brackets, Flanges, and Anchors: Same metal and finish as supported item unless otherwise indicated.
- C. Aluminum Structural Members: 6061-T6 Aluminum
- D. Aluminum Channels: Aluminum Association Standard Channels (AAS)
- E. Aluminum I-Beams: Aluminum Association Standard I-Beams (AAS)
- F. Aluminum Plate: 6061-T6
- G. Aluminum Grating: 1.5" I-Bar Grating, 6063-T6 Aluminum
- H. Rock Bolts: Spin-Lock Type Rock Bolt

2.4 MISCELLANEOUS MATERIALS

- A. Fasteners: Select fasteners of the type, grade, and class required to produce connections capable of withstanding design loadings.
 - 1. For aluminum, provide fasteners fabricated from Type 304 stainless steel and as shown on the Plans and the Construction Notes sheet. All dissimilar metals shall be isolated with Teflon pads, UHMW isolation/bearing pads, nylon washers and/or nylon bushings as identified.

2.5 FABRICATION

- A. Assemble helicopter pad in the shop to greatest extent possible to minimize field splicing and assembly. Disassemble units only as necessary for shipping and handling limitations. Clearly mark units for reassembly and coordinated installation. Use connections that maintain structural value of joined pieces.
- B. Cut, drill, and punch metals cleanly and accurately. Remove burrs and ease edges to a radius of approximately 1/32 inch unless otherwise indicated. Remove sharp or rough areas on exposed surfaces.
- C. Form work true to line and level with accurate angles and surfaces.
- D. Fabricate connections that are exposed to weather in a manner to exclude water. Provide weep holes where water may accumulate.
- E. Cut, reinforce, drill, and tap as needed to receive hardware, screws, and similar items.
- F. Welded Connections: Cope components at connections to provide close fit, or use fittings designed for this purpose. Weld all around at connections, including at fittings.

HELICOPTER PAD RECONSTRUCTION

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 2. Obtain fusion without undercut or overlap.
 3. Remove flux immediately.
 4. Finish exposed welds to comply with NOMMA Guideline 1: "Joint Finishes," for Type 2 welds: All welds are Type 1 welds.
 5. Finish exposed surfaces smooth and blended and, except for fillet welds, so welded surface matches contours of adjoining surfaces.
- G. Form exposed connections with hairline joints, flush and smooth, using concealed fasteners where possible. Where exposed fasteners are required, use Phillips flat-head (countersunk) screws or bolts unless otherwise indicated. Locate joints where least conspicuous.

PART 3 - EXECUTION

3.1 GENERAL

- A. Materials as furnished or fabricated shall result in a strong, tight, well-finished structure.
- B. All materials shall be cut or fabricated accurately to correct measurements.
- C. Construct all items in locations and to elevations indicated on the plans and in conformance with drawings approved by CO under this contract. Prepare subgrades, footings and foundations in accordance with applicable sections of the architect- or engineer-stamped Specifications and Drawings approved under this contract by the CO.
- D. DISPOSAL. Upon completion of work, remove from site any excess building materials including old helicopter pad and associated accessories, and dispose of off Forest Service land in accordance with all local, state, and federal rules and regulations.
- E. Delegated Design:
1. Engage a qualified registered professional engineer, with current State of Alaska professional engineer license, to provide calculations package and completed construction plans.
 2. Structural Performance: The structures shall withstand the effects of gravity loads and the following loads and stresses within limits and under conditions indicated.
 - a. Site specific design requirements for most stringent location shown on the plans including self-weight, snow loads, A-Star and Bell 407 helicopter design loads, static loads, dynamic loads, and rotor loads.
 - b. Lateral Performance: Structures shall withstand the effects of lateral loads as determined by ASCE 7 and as follows:
 - 1) Site specific design criteria for most stringent location shown on the plans for both wind and seismic forces.

HELICOPTER PAD RECONSTRUCTION

3.2 INSTALLATION

- A. Assemble fabricated components to comply with construction drawings, with each component aligned and in correct relation to each other, securely anchored to the supporting column and adjacent structure.
- B. Install helicopter pad accurately in location, alignment, and elevation; level and plumb.
- C. Field Welding is allowed:
 - 1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 - 2. Obtain fusion without undercut or overlap.
 - 3. Remove welding flux immediately.
 - 4. At exposed connections, finish exposed welds and surfaces smooth and blended so no roughness shows after finishing and welded surface matches contours of adjoining surfaces.

3.3 CLEANING AND PROTECTION

- A. Touchup Painting: Immediately after erection, clean field welds, bolted connections, and abraded areas of shop paint and paint exposed areas with same material.

3.4 SURFACE TREATMENTS

- A. Apply surface treatment to telescoping steel post surfaces as indicated on Drawings in a controlled environment prior to shipping to the field. Apply product per manufacturer's recommendations and as directed in Section 09 93 00 – Paint, Stains and Transparent Finishes.

END OF SECTION 13 21 10